

BANKER'S ALGORITHM:

C PROGRAM:

```
#include <stdio.h>

int main()
{
    int n, m, i, j, k;

    int allocation[10][10], max[10][10];
    int need[10][10];
    int available[10];
    int finish[10] = {0};
    int safeSeq[10];
    int count = 0;

    printf("Enter Number of Processes: ");
    scanf("%d", &n);

    printf("Enter Number of Resources: ");
    scanf("%d", &m);

    printf("\nEnter Allocation Matrix:\n");
    for(i = 0; i < n; i++)
        for(j = 0; j < m; j++)
            scanf("%d", &allocation[i][j]);

    printf("\nEnter Maximum Matrix:\n");
    for(i = 0; i < n; i++)
        for(j = 0; j < m; j++)
            scanf("%d", &max[i][j]);

    printf("\nEnter Available Resources:\n");
    for(i = 0; i < m; i++)
        scanf("%d", &available[i]);
```

```

for(i = 0; i < n; i++)
{
    for(j = 0; j < m; j++)
    {
        need[i][j] = max[i][j]- allocation[i][j];
    }
}

while(count < n)
{
    int found = 0;
    for(i = 0; i < n; i++)
    {
        if(finish[i] == 0)
        {
            for(j = 0; j < m; j++)
            {
                if(need[i][j] > available[j])
                    break;
            }
            if(j == m)
            {
                for(k = 0; k < m; k++)
                    available[k] += allocation[i][k];
                safeSeq[count++] = i;
                finish[i] = 1;
                found = 1;
            }
        }
    }
}

```

```

}
if(found == 0)
{
printf("\nSystem is NOT in Safe State\n");
return 0;
}
}

printf("\nSystem is in Safe State\n");
printf("Safe Sequence: ");
for(i = 0; i < n; i++)
printf("P%d ", safeSeq[i]);
printf("\n");
return 0;
}

```

SHELL SCRIPTING:

```

#!/bin/bash

echo "Enter Number of Processes:"

read n

echo "Enter Safe Sequence (space separated):"

read -a seq

echo "Safe Sequence is:"

for ((i=0;i<n;i++))

do

echo -n "P${seq[$i]} "

done

echo

echo "System is in Safe State"

```

OUTPUT:

```
(kali㉿ Anbuchelvan)-[~]  
$ nano banker.sh
```

```
(kali㉿ Anbuchelvan)-[~]  
$ chmod +x banker.sh
```

```
(kali㉿ Anbuchelvan)-[~]  
$ bash banker.sh
```

Enter Number of Processes:

2

Enter Safe Sequence (space separated):

3

Safe Sequence is:

P3 P

System is in Safe State